

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2022

Course Code: EIE101R01

Course: BASIC ELECTRONICS ENGINEERING

Question Paper No. : UGF010

Max. Marks:100

PART – A

Answer all the questions

10 x 2 = 20 Marks

1. Explain the depletion layer formation in the pn junction diode.
2. Compare the breakdown mechanism in pn junction and Zener diodes.
3. Construct the clamper circuit to clamp the input voltage of $v = 6\sin\omega t$ and plot its input-output waveform.
4. Distinguish the basic principle of operation of LED and photodiodes.
5. Evaluate β and γ for the current amplification factor of CB connection is 0.96.
6. Summarize the key factors for achieving faithful amplification in BJT.
7. Why Base resistor bias is called fixed bias?
8. Compare the transfer characteristics of JFET, D MOSFET, and E MOSFET.
9. A power supply has a voltage regulation of 1%. If the no-load voltage is 40V, what is the full-load voltage?

10. Determine the unknown frequency of the input signal applied to the vertical deflection plate when the horizontal deflection plate frequency is 2kHz for the Lissajous pattern obtained, as shown below.



PART - B

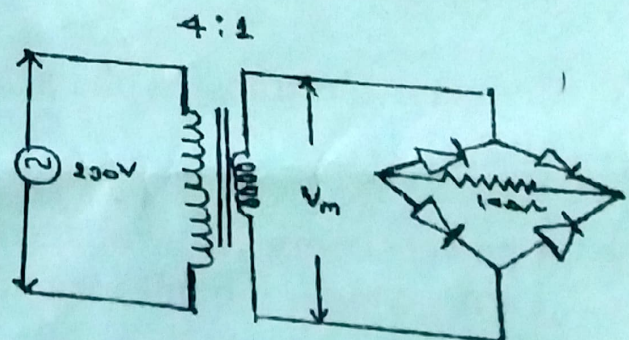
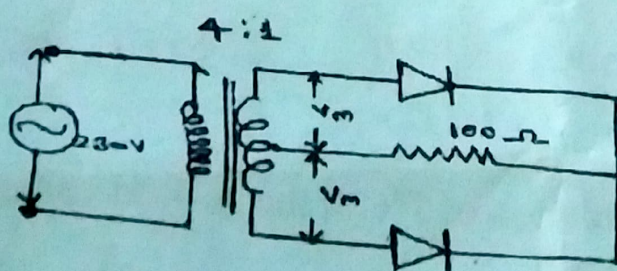
Answer all the questions

4 x 15 = 60 Marks

11. (a) The applied input voltage (E_i) to the Zener voltage regulator is 16V, and the value of the series and load resistors are $1K\Omega$ and $1.2K\Omega$. Determine the regulated output voltage, drop across the series resistor, the current through Zener and Zener power dissipation for $V_z = 10V$, and Maximum power dissipation of Zener diode is 30mW. (6)
- (b) Explain with necessary illustration construction and working of Zener diode. (9)

(OR)

12. The center-tap and bridge type circuits are shown to have the same load resistance and transformer turn ratio. The primary of each is connected to a 230V, 50Hz supply. Find (i) the dc voltage in each case (ii) PIV for each for the same d.c output. Assume the diodes to be ideal.

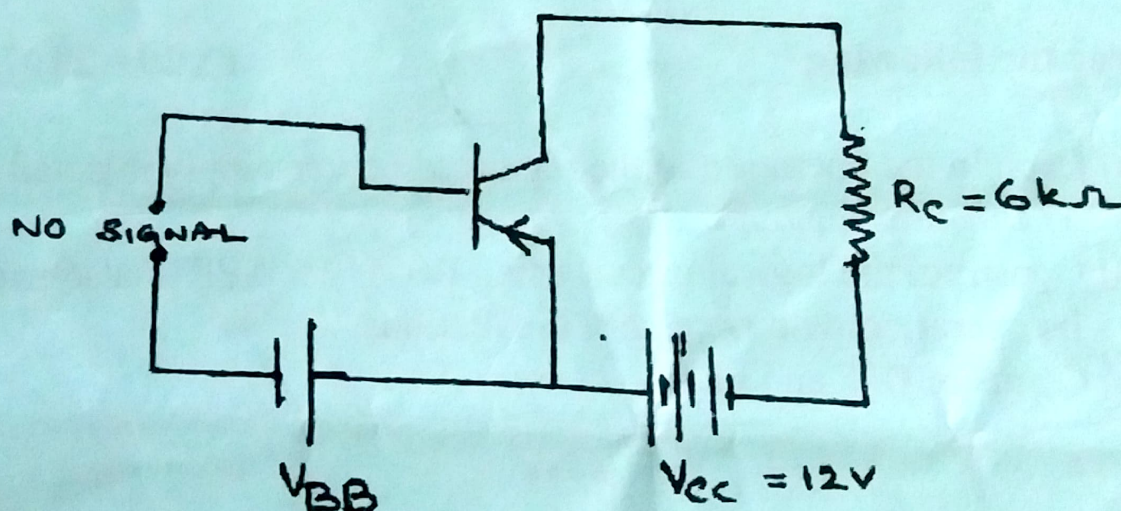


13. (a) Explain input and output characteristics of NPN transistor in common emitter configuration with neat diagram and characteristics. (10)

(b) A transistor is connected in a common emitter configuration in which the collector supply is 8V, and the voltage drop across resistance R_c connected in the collector circuit is 0.5V. The value of $R_c = 700\Omega$. If $\alpha = 0.9$ determine (i) Collector-emitter voltage. (ii) Base current. (5)

(OR)

14. (a) Explain the importance of the d.c load line. Also, explain how you will draw a d.c load line on the output characteristics. If $V_{cc}=12V$ and $R_c=6K\Omega$, draw the d.c load line in the circuit shown. What will be the Q-point if zero signal base current is $20\mu A$ and $\beta = 50$? (11)



(b) Differentiate transistor in CB, CE, and CC configuration. (4)

15. Explain with necessary illustrations the construction and working of n channel JFET. Also, draw its output and transfer characteristics.

(OR)

16. As a Hardware Engineer in an integrated circuit (IC) manufacturing company, your team leader provides you with a transistor with the following property. It acts as a voltage-controlled device. This device has n-type material that has been defused into a p-type substrate, and

the free electrons flow from the source to drain through the narrow channel between the gate and p-type region. Identify that semiconductor device provided to you. Also, explain the construction, working, and characteristics with a neat diagram.

17. (a) With the neat schematic, explain the working principle of the cathode ray oscilloscope in detail. (10)
(b) Explain in detail how frequency and voltage measurements are done using an Oscilloscope. (5)

(OR)

18. Discuss the importance of voltage regulation in power supplies. Also, explain with a neat diagram the transistor shunt voltage regulator.

PART – C

Answer the following

1 x 20 = 20 Marks

19. (a) Explain the operation of the regulated power supply in detail with a neat block diagram. (10)
(b) Construct the logical circuit using two E MOSFETs also explain its operation with necessary illustrations. (5)
(c) Compare BJT and FET. (5)
